

JARED (Just A Really Expensive Device)

Master Architecture & Development Plan

Project Vision

- Fully local, privacy-first assistant and device control platform.
- Commercial-grade architecture from day one.
- No cloud accounts, no telemetry, no hidden data transfer.
- Scalable from single-car garage to full shop.
- Supports voice control, device automation, manuals library, and project memory.

Core System Pillars

- Audio Layer (Wake word, STT, TTS, ducking, barge-in).
- NLP & Context Layer (Intents, multi-turn context, decision capture).
- Knowledge Layer (Manual ingestion, searchable library, offline-first routing).
- Device Platform (Registry, templates, policy engine, MQTT local control).
- Security & Policy Engine (Class 0/1/2 safety rules, PIN auth).
- Observability (Logging, diagnostics, crash recovery).

Development Milestones

Milestone 0 – Skeleton & Offline Guardrails

- Scaffold repository structure.
- Implement strict local-only network policy gate.
- Add structured logging system.
- Confirm system boots and runs in offline mode.

Milestone 1 – Audio Foundation

- Device input/output selection.
- Level metering and calibration.
- Music playback routing with ducking slider.
- Persistent configuration storage.

Milestone 2 – Voice Pipeline

- Wake word detection engine abstraction.
- Local speech-to-text integration.
- Basic intent parsing (10–20 starter commands).
- Event bus integration.

Milestone 3 – Device Platform

- Device registry (rooms, devices, templates).
- ActionRequest schema and Policy Engine (Class 0/1/2).
- PIN phrase authentication + authorization window.
- Local MQTT driver.
- End-to-end allow/deny device control flow.

Milestone 4 – UI Command Center

- Dashboard with mic activity and status.
- Settings panel (ducking, wake mode, offline toggle).
- Device control view.
- Review queue for decisions and fixes.

Milestone 5 – Memory & Troubleshooting Intelligence

- Decision capture (hybrid approval model).
- ‘That worked’ troubleshooting capture system.
- Searchable project memory.
- Project-scoped knowledge separation.

Milestone 6 – Manuals Library

- PDF ingestion (text + OCR).
- Keyword index (semantic later).
- Procedure-style answer composer.
- Hidden provenance with on-demand page lookup.

Milestone 7 – Commercial Polish

- Barge-in stability improvements.
- Calibration wizard.
- Device verification (confirmed off behavior).
- Diagnostics export bundle.
- Service-mode auto start.

Safety Model Overview

- Class 0 – Safe actions (lights, music).
- Class 1 – Caution actions (general outlets).
- Class 2 – Critical actions (heater ON, compressor ON, garage door).
- Class 2 ON requires PIN or PTT authorization.
- Class 2 OFF does not require authorization but confirms state.

Knowledge Routing Policy

- Manuals-first for project-specific procedural questions.
- Project memory for build decisions.
- Offline reference pack for definitions.
- Online search only with explicit user permission.
- Manuals never uploaded externally.

Long-Term Expansion Points

- Multi-room mic arbitration.
- Voice identity recognition.
- Automation rule engine.
- Marketplace pack system (voices, wake words, skills).
- Local-only mobile interface.
- Hardware controller nodes (ESP32, relays).

Operating Principles

- Accuracy over ego – system allowed to say 'I'm not sure.'
- No hallucinated specs or procedures.
- Transparent confidence and escalation.
- All critical actions logged.
- Local-first architecture always preserved.